

JAMES DAY

Games Programmer

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Battle, UK jambient.github.io

PERSONAL STATEMENT

A Games Programmer with a First-Class BSc (Hons) in Computer Games Programming and industry experience at Wargaming. I am a strong C++ developer with knowledge of engine, rendering, and gameplay systems with projects involving custom ECS architectures, rendering frameworks, asset pipelines, physics simulations, and editor tooling.

EXPERIENCE

Software Engineering Intern (Wargaming, UE5 C++)

06/2025 - 08/2025 Guildford, UK

- Designed and implemented whole gameplay systems into prototype.
- Developed custom asset editor tools using Unreal Slate to improve workflows for designers. Maintained constant communication for feedback with design team to keep goals clear.
- Researched and evaluated benefits for various gameplay frameworks (such as GAS), and contributed insights during team discussions.

RELEVANT PROJECTS

Quack Engine (C++, 2026)

- Developed rendering systems for a cross-platform game engine, including a custom Rendering Hardware Interface (RHI) layer abstracting away Windows and PS5 graphics libraries.
- Proved RHI capabilities by creating a single ImGui backend that supports all platforms.
- Designed and implemented a two-stage asset pipeline with offline processing and runtime asset management.
- Implemented MSDF text rendering for high-quality scalable text.
- Collaborated within a 5 person team, integrating my rendering and asset systems into the architecture.

SDF Modelling Engine (C++)

- Built a Signed Distance Field (SDF) modelling engine with type-registry system, scene graph, and compilation into GPU readable data.
- Developed a ImGui editor interface for scene modification including serialisation and material management.

3D ECS Physics Engine (C++, 2025)

- Developed an archetype-based Entity Component System (ECS) optimised for cache-efficient data access.
- Built an impulse-based rigid body physics solver using iterative constraint resolution
- Integrated DirectX 11 rendering to visualize the physics engine with ImGui for debugging.

KEY SKILLS

Programming Languages

C++, C#, Python, SQL, Lua

Game Engines

Unreal Engine 5, Unity

Frameworks

DirectX 11 & 12, OpenGL, SDL2, MonoGame

Version Control

Git, GitHub, Perforce

Web Technologies

HTML, CSS, JavaScript

Soft Skills

Problem Solving, Teamwork

HOBBIES & INTERESTS

Piano

I have played piano since the age of 7 and enjoy learning new pieces in my spare time.

Composition

I produced and released various tracks on SoundCloud, with one song receiving over 20,000 plays.

UI/UX Design

I enjoy designing both website and game UI/UX. This can be seen in quite a few of my projects primarily Astro Forge and Tanks.

EDUCATION & AWARDS

TIGA Awards Nominee

For the UK Games Education Awards 2026

University Graduate Exhibition - First Place

My project *SDF Modelling Engine* was voted the best project by industry experts

University of Staffordshire, (2023 - 2026)

Computer Games Programming BSc (Hons) - First Class Honours

Battle Abbey School, (2018 - 2023)

9 GCSEs and 3 A levels comprising: A in Computer Science and Mathematics and a C in Economics.